# Zane Beeai

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## Education

**University of Waterloo** — Waterloo, ON
*B.A.Sc. in Biomedical Engineering* — *Class of 2030*
**Coursework:** Calculus I & II, Multivariable Calculus, Linear Algebra, Differential Equations, Physics I & II

## Technical Skills

**Languages**: Java, Python, Javascript, C#, C++, VHDL, Visual Basic, R, Matlab
**Frameworks**: React, Node.js, MongoDB, LangChain
**Libraries**: PyTorch, TensorFlow, OpenCV, Pinecone, LangChain, NumPy, SymPy, pandas, Matplotlib

## Experience

**Co-Founder** — Jan 2025 – June 2025
*Contractual (VC-Backed Startup)* — *Toronto, ON*

- Designed and deployed a **retrieval-augmented generation (RAG)** and **GraphRAG** backend using **Pinecone**, **LangChain**, and custom text chunking pipelines to automate tender analysis for contractors.
- Led the development of a **matching engine** and **searchable database**, integrating **NLP-driven** insights with a guided UX to streamline public tender discovery and bidding.
- Helped contractors secure **$600,000+ in Canadian government contracts**, driving measurable ROI and informing the company's **go-to-market strategy** with early pilot users.

**Research Assistant** — July 2024 – June 2025
*Sunnybrook Research Institute (FUS Lab)* — *Toronto, ON*

- Researched microbubble cavitation in agarose vessels using simultaneous **optical and acoustic data** to study antivascular therapy applications under Dr. David Goertz.
- Developed a custom **MATLAB GUI** for real-time **signal processing** and image segmentation, cutting manual analysis time by **80%** and enabling high-throughput experimentation.
- Implemented **SOTA object tracking models** and novel **time-frequency transformations** to analyze micron-scale bubble dynamics, improving insight into therapeutic ultrasound mechanisms.

**Founder** — Sept 2023 – April 2025
*WOSS Robotics 8433* — *Oakville, ON*

- Founded and scaled a competitive **high school robotics program**, mentoring **3 teams** through international-level competitions and achieving a **Top 100 global skills ranking**.
- Secured over **$14,000 in sponsorships** through partnerships with businesses and STEM organizations, funding expansion into advanced mechanical design, coding, and outreach initiatives, featured on **Oakville News**.
- Led teams to earn **provincial, national, and international awards** including **Excellence** and **Skills Ontario**, culminating in qualification for the **VEX World Championships 2025**.

## Projects

**Solara CFC** | *Python, RASTER, JS, React, Flask, Next.js* — /Solara_CFC

- Top 40 teams/56,000 participants in NASA Space Apps Challenge 2023. Pulled data from NASA & CSA satellites of CFC emissions, solar flux, and ozone to overlay on a mapping frontend, improving NASA's VISIONS software.

**WarRig X1 - Autonomous EV** | *CAD, FEA, Circuit Design, Python, LiDAR, Computer Vision* — /WarRig

- Designed and built 24V electric race car from scratch. Awarded Dennis Weishar Engineering Design Award at uWaterloo's EV Challenge. Funded by Town of Oakville for $12,000 to make the car self-driving with LiDAR.

## Research

**TraceGR: Modular Spacetime Imaging using Null Geodesics in Raycasting**
doi.org/10.1109/URTC65039.2024.10937597 — University of Toronto (2024)
**Zane Beeai**, Aditya Makkar, Dr. Ahmed Ellithy
Led research under Dr. Ahmed Ellithy to create TraceGR: a null geodesic-based, modular raycaster to produce spacetime images rivaled only by Christopher Nolan in Interstellar. Presented at **Canadian Undergraduate Math Conference (CUMC)** and **MIT-IEEE URTC**; peer-reviewed *IEEE Xplore* publication.